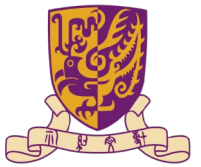


香港中文大學
The Chinese University of Hong Kong

# 合见工软产学融合赛题
# Thermal-aware 3D Floorplanning Placer
# An Optimization Solution Based on FM-ILP Hybrid Algorithm

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The Chinese University of Hong Kong 香港中文大学
2025-10-11

# Outline

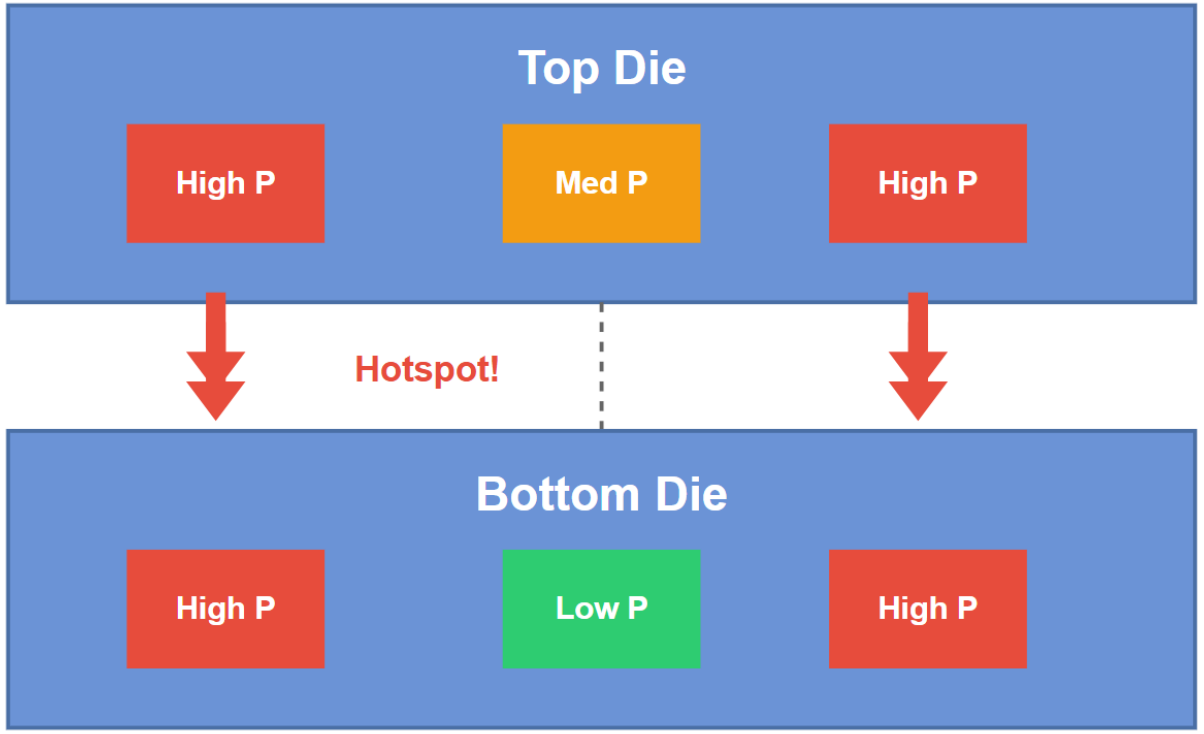


- Background
- Formulation
- Algorithm Design
- Experiments

# Thermal Challenges in 3D ICs

- 3D stacking extends Moore‘s Law;
- Vertical integration improves performance and density;
- TSV or hybrid bump interconnects enable inter-layer communication

- Vertical heat conduction;
- Cross-layer thermal coupling;
- Thermal impact – warpage etc.



# Formulation

Input:
- Blocks Netlist
- Power

Output:
- Die partition
- 3D layout

Objective:
- Co-optimize the thermal and wirelength

**Objective Function**

*minimize:* $\alpha \cdot \text{HPWL} + \beta \cdot \text{Thermal_Penalty}$

$\text{HPWL} = \Sigma\, \text{weight}_{ij} \cdot (|x_i - x_j| + |y_i - y_j|)$

**Decision Variables**

- $x_i, y_i \in [0, W] \times [0, H]$ (module coordinates)
- $r_i \in \{0, 1\}$ (rotation flag)

**Constraints**

√ Boundary: $x_i \geq 0.1\text{mm}$

√ Non-overlap (same layer)

√ Min spacing: 100μm

√ Thermal: $z_{ij}$ constraints

**Thermal Penalty Components**

**Intra-layer**

- Same die, high power
- Distance < threshold
- Penalty $\propto (P_i + P_j)$

**Cross-layer**

- Top ↔ Bottom
- $z_{ij} = 1$ if overlap
- Penalty $= (P_i + P_j) \cdot z_{ij}$

# Algorithm Design

**Objective:** $\min \alpha \cdot \sum_{(i,j)\in\mathcal{E}} C_{ij} \cdot (|c^x_i - c^x_j| + |c^y_i - c^y_j|) + \beta \cdot \Phi_{thermal} + \gamma \cdot \Delta P$

## Decision Variables

**Position:**

$x_i, y_i \in [0, W] \times [0, H]$ (continuous)

**Die Assignment:**

$z_i \in \{0, 1\}$ (0=bottom, 1=top)

**Rotation:**

$r_i \in \{0, 1\}$ (0=H, 1=V, for non-square)

## Key Definitions

**Module Center:**

$c_i = (x_i + w^{eff}_i/2, y_i + h^{eff}_i/2)$

**Effective Dimensions:**

$w^{eff}_i = w_i \cdot (1 - r_i) + h_i \cdot r_i$

$h^{eff}_i = h_i \cdot (1 - r_i) + w_i \cdot r_i$

**Connection Strength:**

$C_{ij}$: wirelength weight between *i* and *j*

## Penalty Terms

- $\Phi_{thermal}$: Thermal penalty accounting for:
  - Intra-layer hotspots (high-power modules too close on same die)
  - Cross-layer thermal coupling (overlapping high-power modules)
- $\Delta P$: Inter-layer power imbalance
  - Balances power distribution between top and bottom dies

# Algorithm Design

**Algorithm 1** ILP-Greedy Hybrid Solver

**Require:** Thermal blocks $\mathcal{B}$, region size $(W, H)$, simple nets $\mathcal{N}_s$, detailed nets

**Ensure:** Block locations, orientations, die assignment

$T_1 \leftarrow 0.95 \times T$, $T_2 \leftarrow 0.05 \times T$

// **Phase 1: Global ILP Optimization**

$(\mathcal{L}, \mathcal{O}_c, \mathcal{D}) \leftarrow$ PHASE1-ILP$(\mathcal{B}, \mathcal{N}_s, W, H, T_1)$

// $\mathcal{L}$: locations, $\mathcal{O}_c$: coarse orientations (H/V), $\mathcal{D}$: die assignment

**if** detailed nets available **and** $T_2 > 0$ **then**

  // **Phase 2: Pin-Aware Greedy Refinement**

  $\mathcal{O}_f \leftarrow$ PHASE2-GREEDY$(\mathcal{B}, \mathcal{L}, \mathcal{O}_c, \mathcal{N}_d, T_2)$

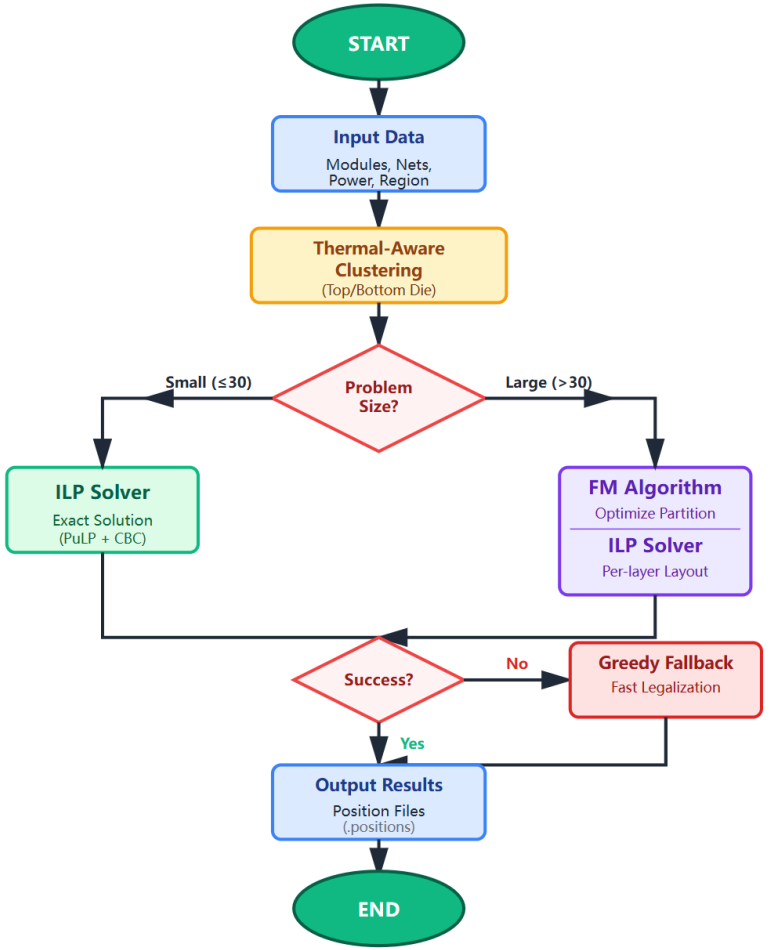
  // $\mathcal{O}_f$: fine-grained orientations (R0/R90/R180/R270)

**else**

  $\mathcal{O}_f \leftarrow \mathcal{O}_c$

**end if**

**return** $(\mathcal{L}, \mathcal{O}_f, \mathcal{D})$ =0



# Constraints – Cross-Layer Relationship Variable

Introduce layer discontinuity variable $d_{ij}$ (indicates *i* and *j* are on different layers):

$$d_{ij} \geq z_i - z_j$$

$$d_{ij} \geq z_j - z_i$$

$$d_{ij} \leq z_i + z_j$$

$$d_{ij} \leq 2 - z_i - z_j$$

Define same-layer indicator: $s_{ij} = 1 - d_{ij}$

When same layer, enforce minimum spacing in at least one direction:

$$\ell_{ij} + r_{ij} + b_{ij} + a_{ij} \geq s_{ij}$$

# Constraints – Directional Separation

**Same-layer non-overlap constraints** (e.g., left side):

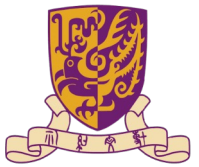
$$x_i + w^{\text{eff}}_i + \delta_{\min} \le x_j + M \cdot (1 - \ell_{ij})$$

where $\delta_{\min}$ = 0.1 mm (100 μm minimum spacing),
$M$ is a sufficiently large constant (big-M method).

**Similar constraints for other directions:**

- Right: $x_j + w^{\text{eff}}_j + \delta_{\min} \le x_i + M \cdot (1 - r_{ij})$
- Bottom: $y_i + h^{\text{eff}}_i + \delta_{\min} \le y_j + M \cdot (1 - b_{ij})$
- Above: $y_j + h^{\text{eff}}_j + \delta_{\min} \le y_i + M \cdot (1 - a_{ij})$

# Constraints – Thermal Penalty



For high-power modules ($P_i$, $P_j \geq P_{threshold}$), introduce cross-layer overlap penalty. Define auxiliary variable $o_{ij} \in \{0, 1\}$:

$$\ell^x_{ij} + r^x_{ij} + b^z_{ij} + a^z_{ij} \geq d_{ij} - o_{ij}$$

**Thermal penalty:**

$$\Phi_{thermal} = \sum_{\substack{i \in top, j \in bottom \\ P_i, P_j \geq P_{threshold}}} (P_i + P_j) \cdot o_{ij} + \Phi_{same\text{-}layer}$$

where $\Phi_{same\text{-}layer}$ accounts for intra-layer high-power modules that are too close (distance below threshold).

# Constraints – Inter-Layer Balance

$$P_{\text{bottom}} = \sum_{i=1}^{n} P_i \cdot (1 - z_i)$$

$$P_{\text{top}} = \sum_{i=1}^{n} P_i \cdot z_i$$

$$\Delta P = |P_{\text{bottom}} - P_{\text{top}}|$$

**Linearization using absolute value:**

$\Delta P = p^+ + p^-$, where

$P_{\text{bottom}} - P_{\text{top}} = p^+ - p^-$

($p^+$, $p^- \geq 0$ are auxiliary variables)

# Algorithm Design

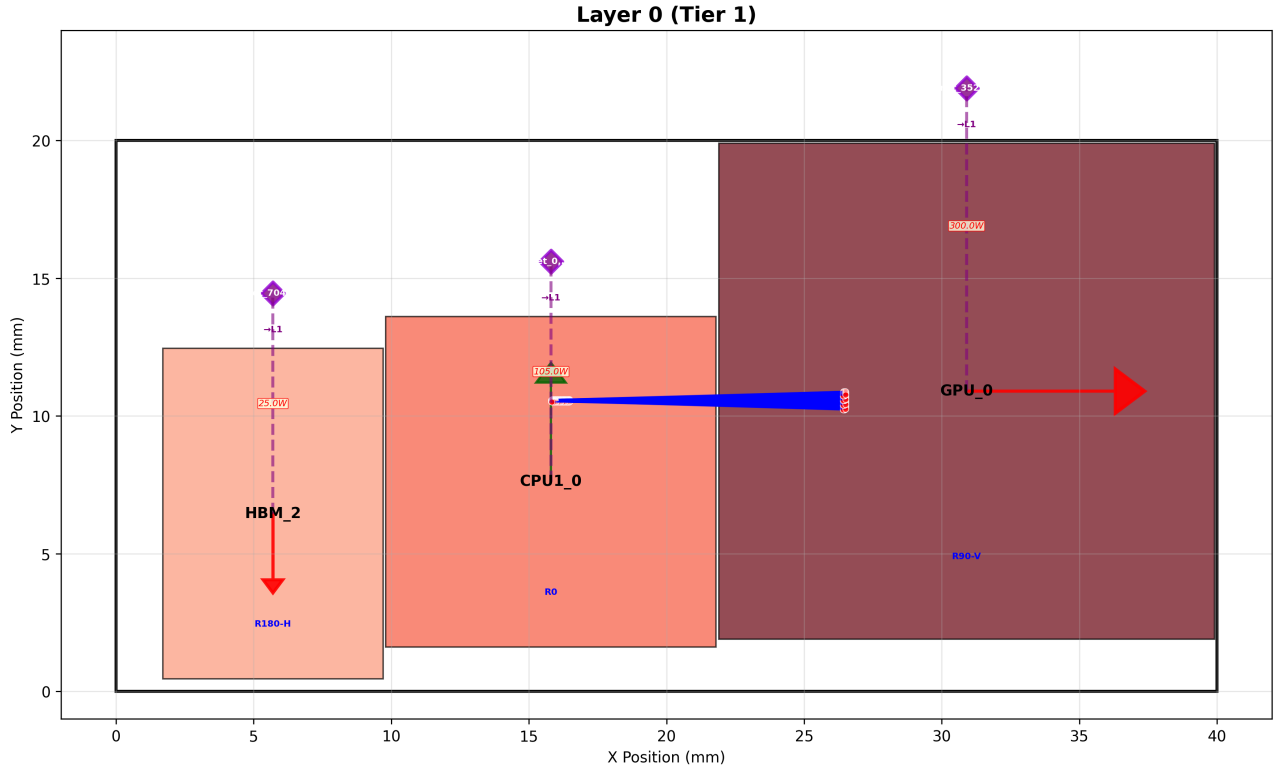
**Algorithm 2** Phase 2: Pin-Aware Greedy Refinement

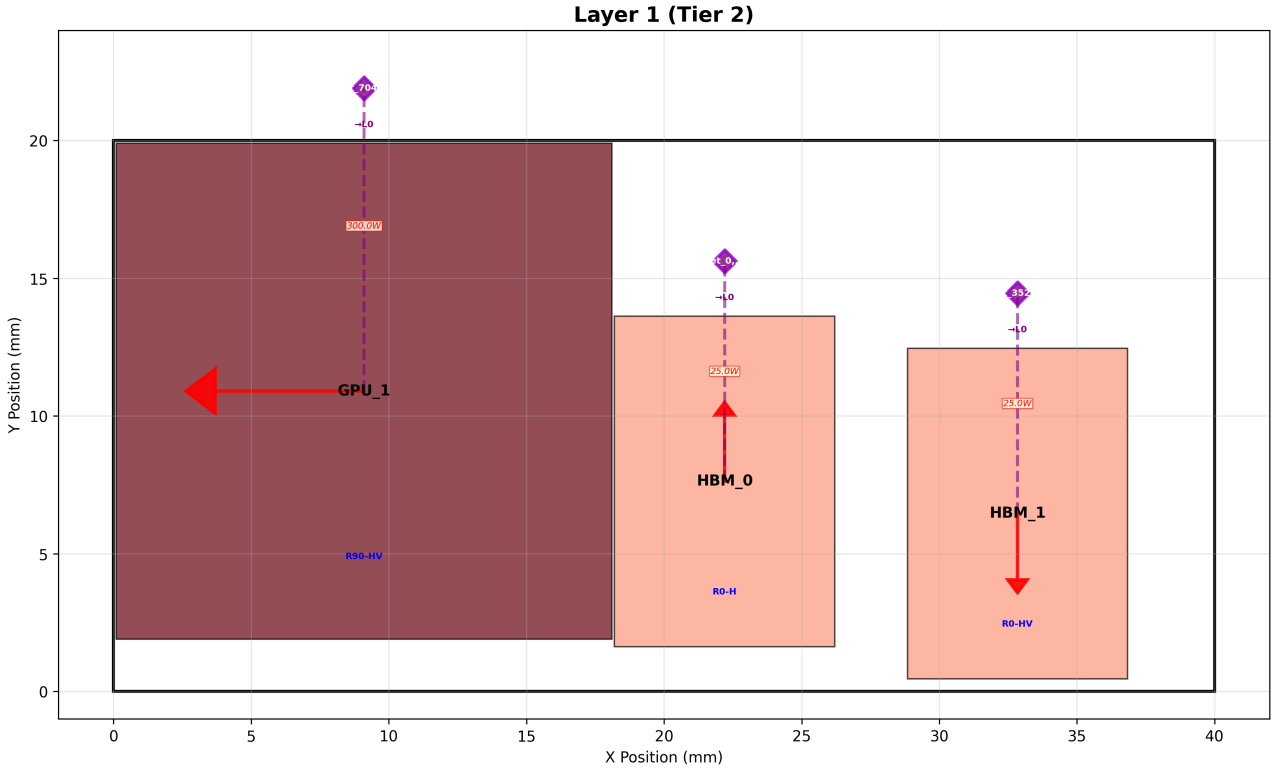
**Require:** Locations $\mathcal{L}$, coarse orientations $\mathcal{O}_c$, detailed nets $\mathcal{N}_d$, timeout $T_2$
**Ensure:** Fine-grained orientations $\mathcal{O}_f$

$\mathcal{O}_f \leftarrow \mathcal{O}_c$ // Initialize with Phase 1 result
$t_{\text{start}} \leftarrow$ current_time()
**for** each block $i$ in $\mathcal{B}$ **do**
  **if** current_time() $-t_{\text{start}} > T_2$ **then**
    **break** // Timeout
  **end if**
  // Determine allowed rotations based on footprint
  **if** $i$ is square **then**
    $\mathcal{R}_i \leftarrow$ {R0, R90, R180, R270}
  **else if** $\mathcal{O}_c[i] =$ horizontal **then**
    $\mathcal{R}_i \leftarrow$ {R0, R180} // Preserve $w \times h$ footprint
  **else**
    $\mathcal{R}_i \leftarrow$ {R90, R270} // Preserve $h \times w$ footprint
  **end if**
  // Greedy selection based on pin distance
  $r^* \leftarrow \arg\min_{r \in \mathcal{R}_i} C_{\text{pin}}(i, r)$ // Eq. (23)
  $\mathcal{O}_f[i] \leftarrow r^*$
**end for**
**return** $\mathcal{O}_f$ =0

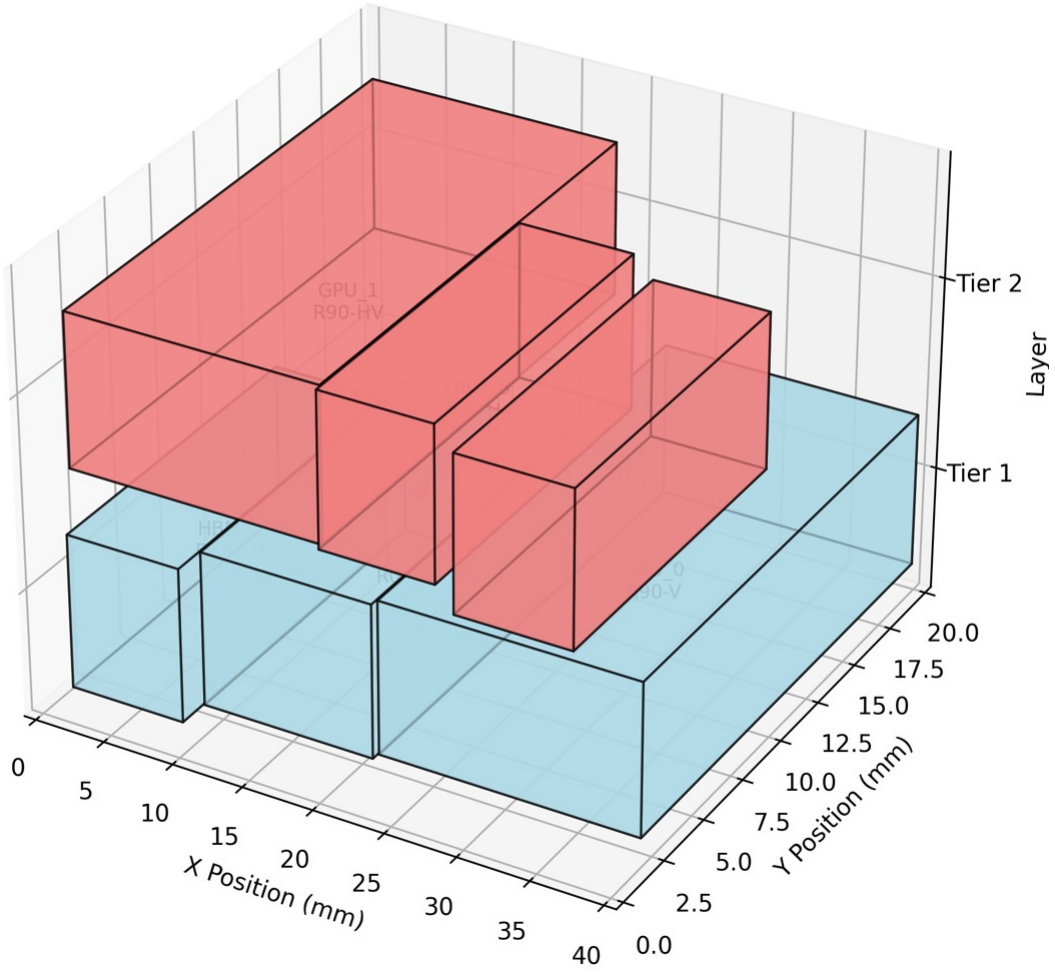
$$r_i^* = \arg\min_{r \in \mathcal{R}_i} C_{\text{pin}}(i, r) \tag{23}$$

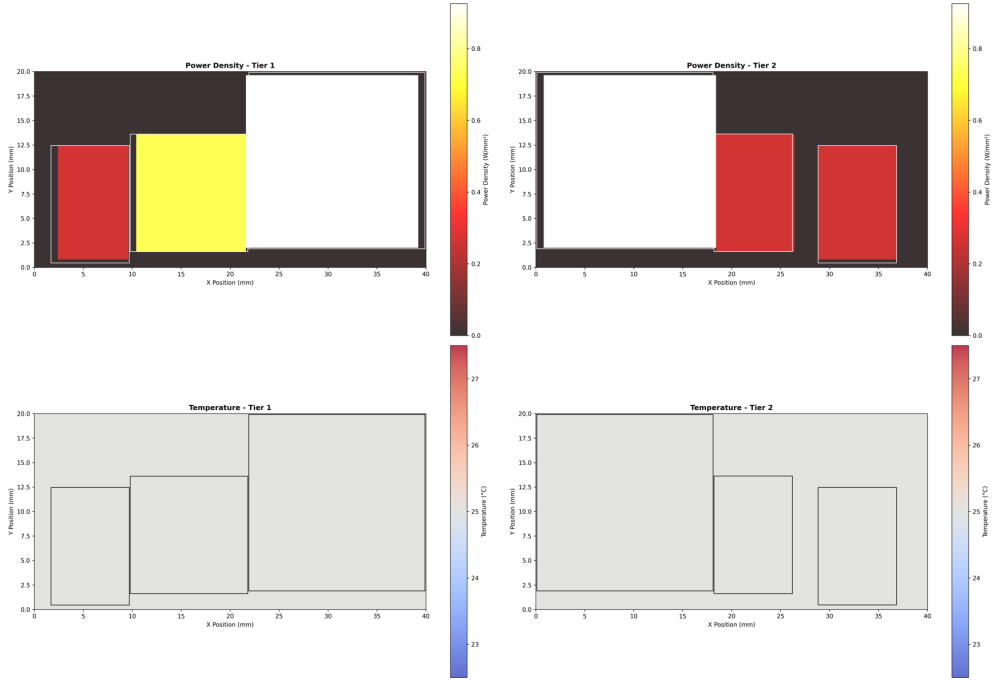
# Placer Experiments – Case 1



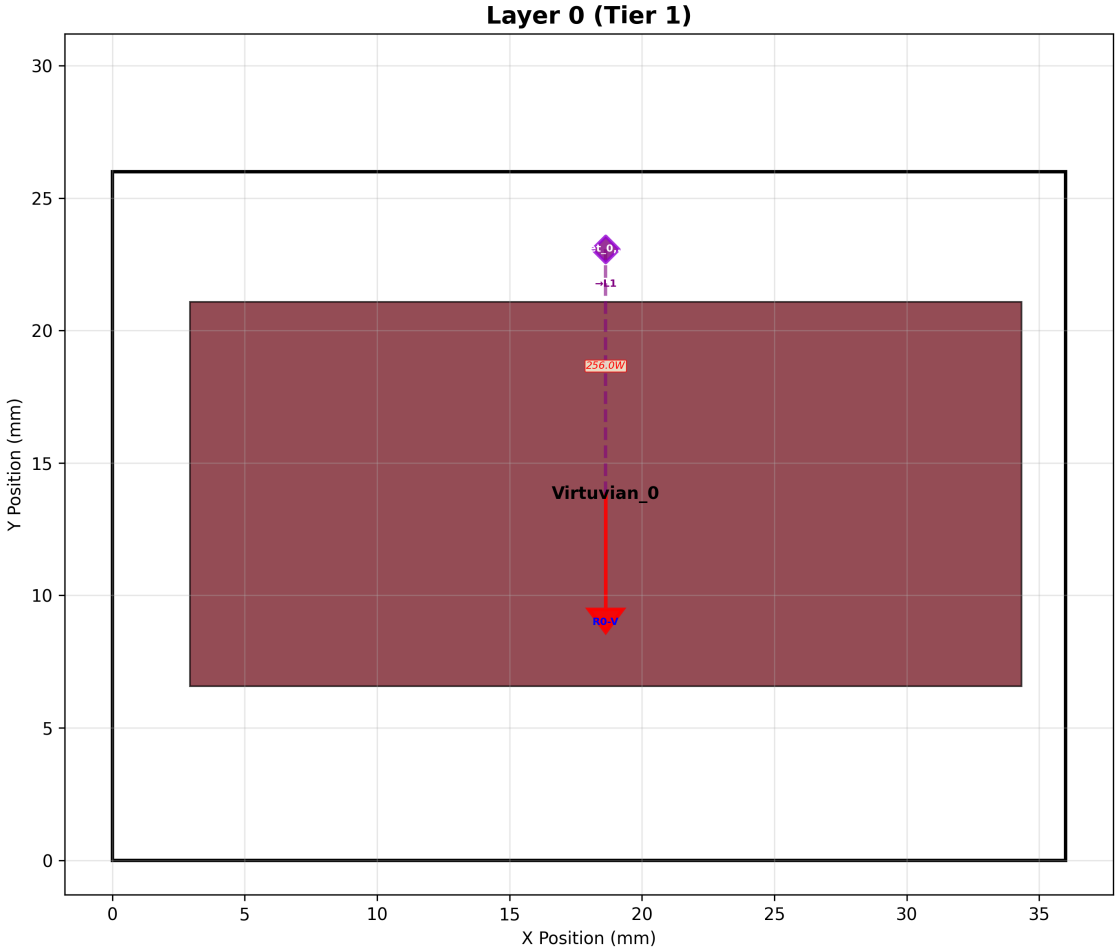
# Placer Experiments – Case 1

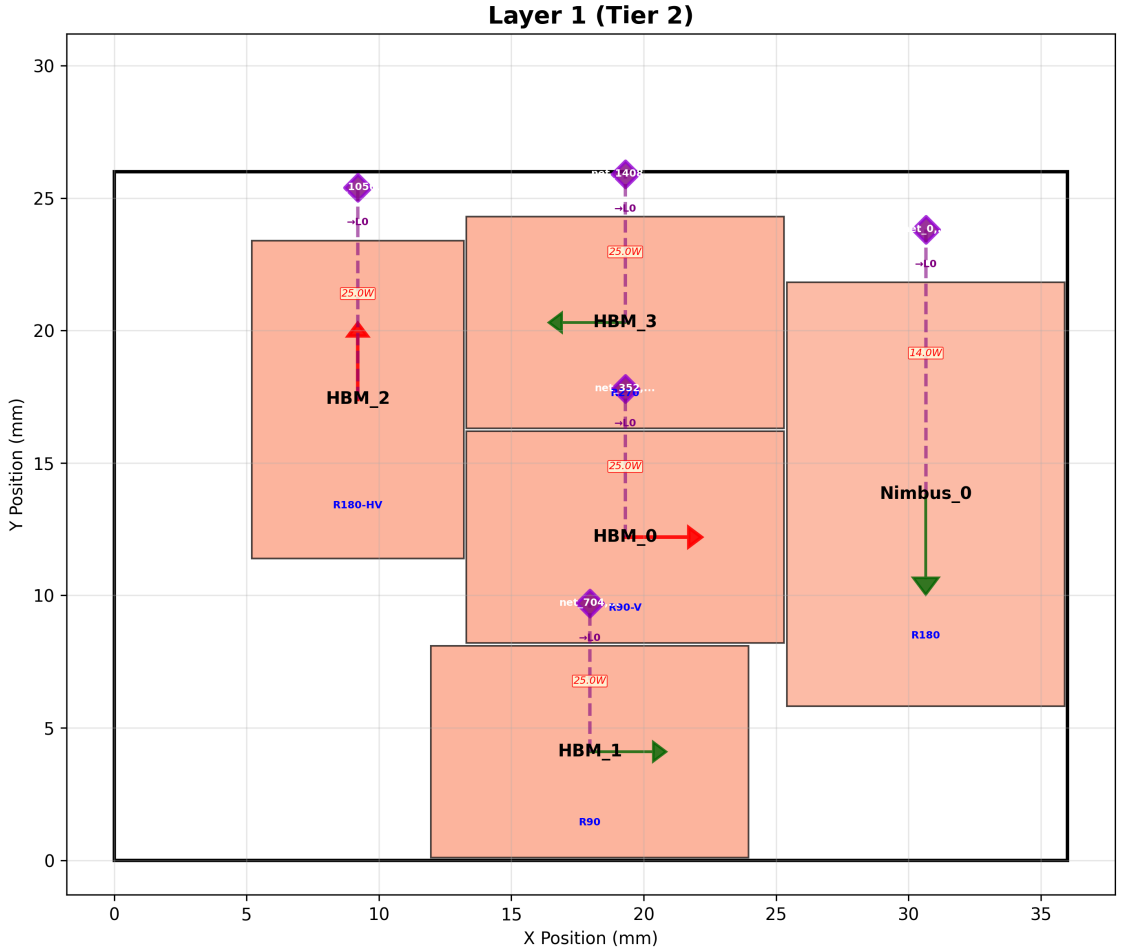


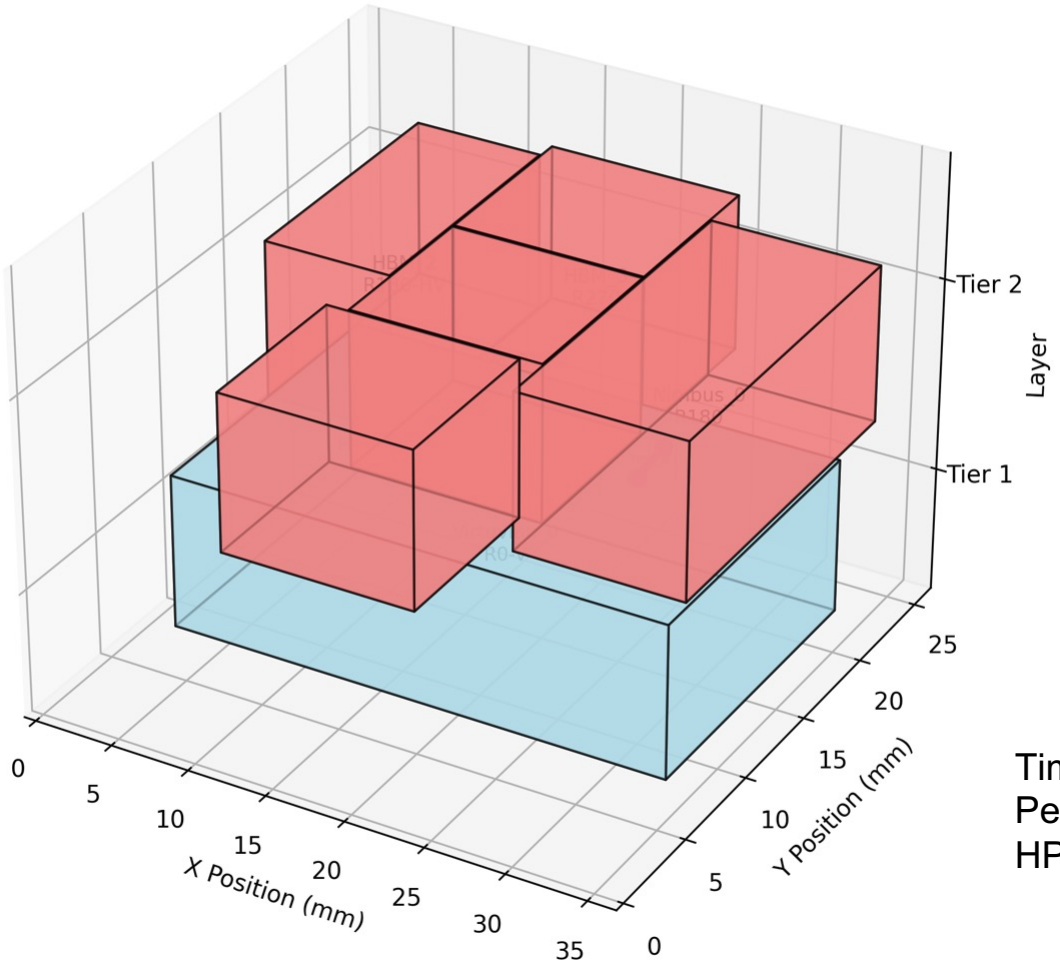

Time: 15.891757011413574
Peak Temperature: 306.103
HPWL: 5936.29 mm

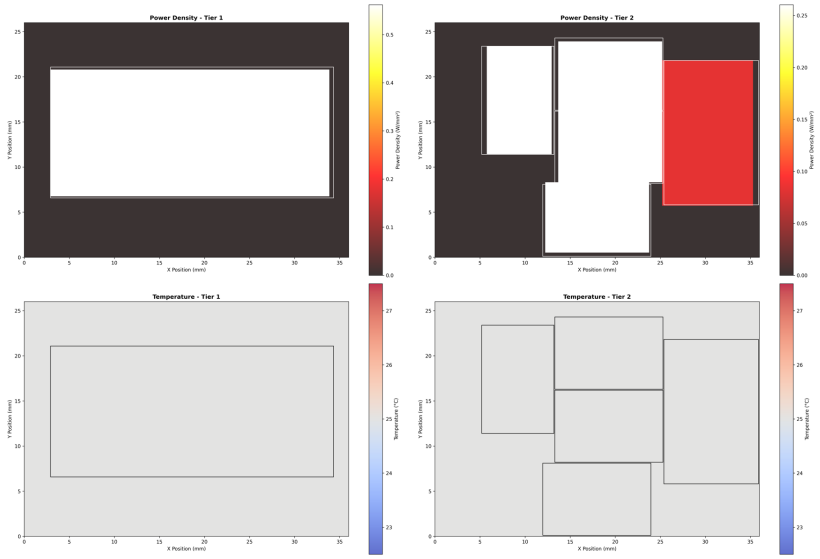
# Placer Experiments – Case 2



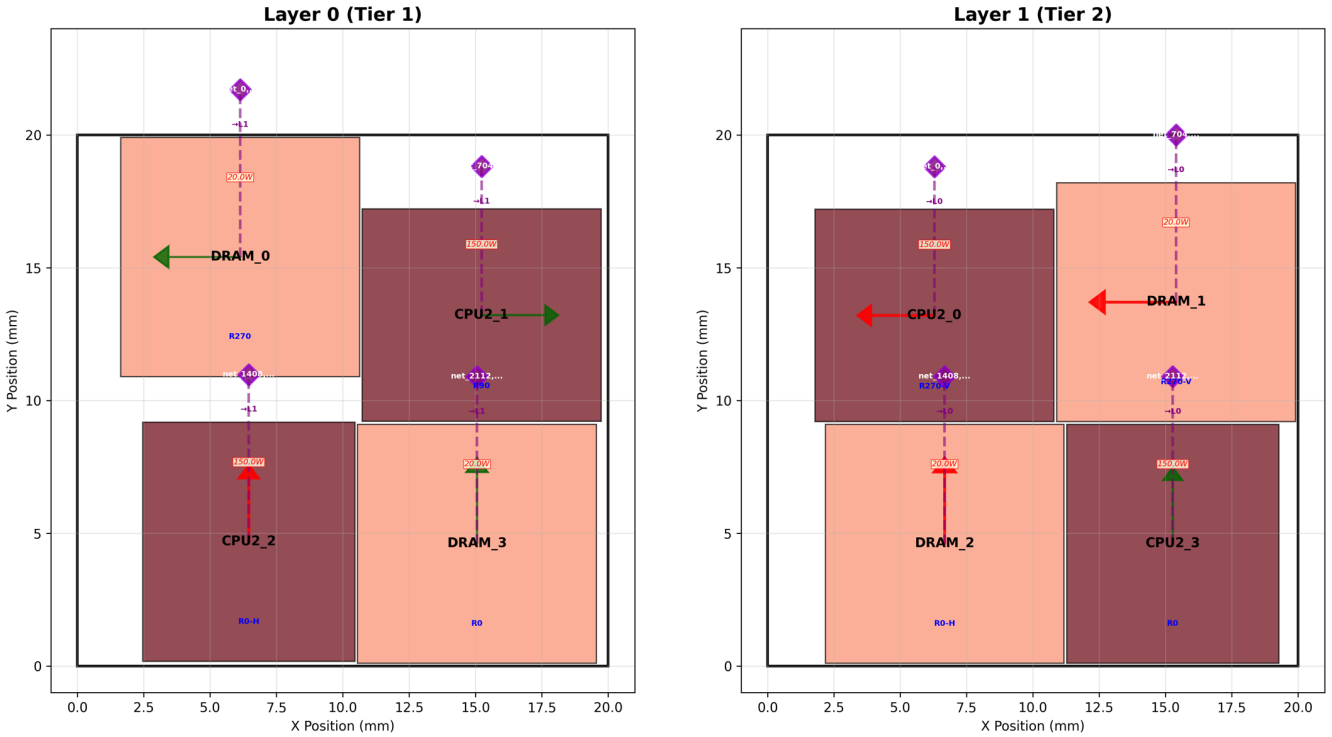
# Placer Experiments – Case 2



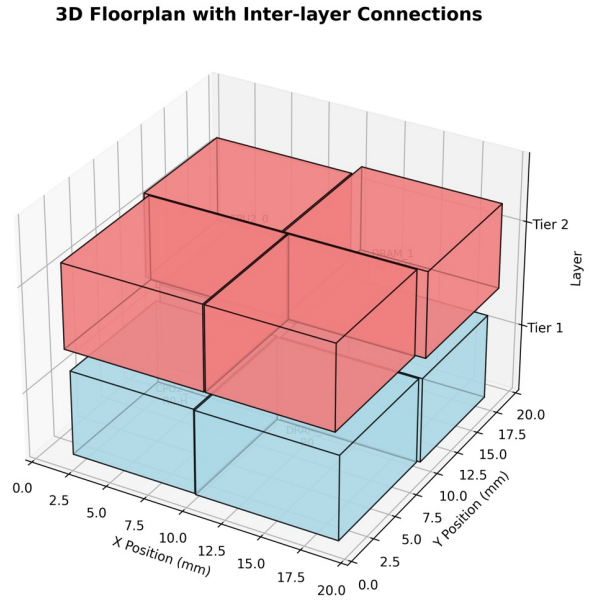

Time: 111.38
Peak Temperature: 306.05
HPWL: 5054.79 mm

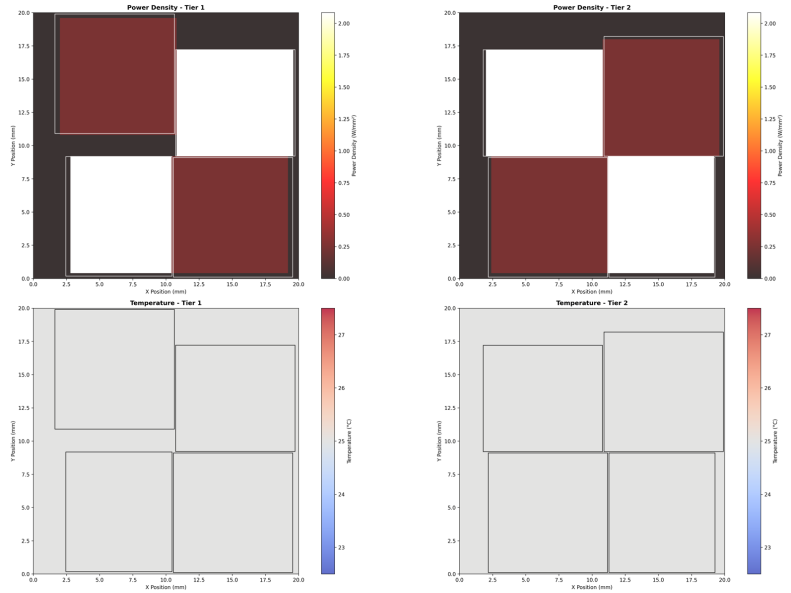
# Placer Experiments – Case 3





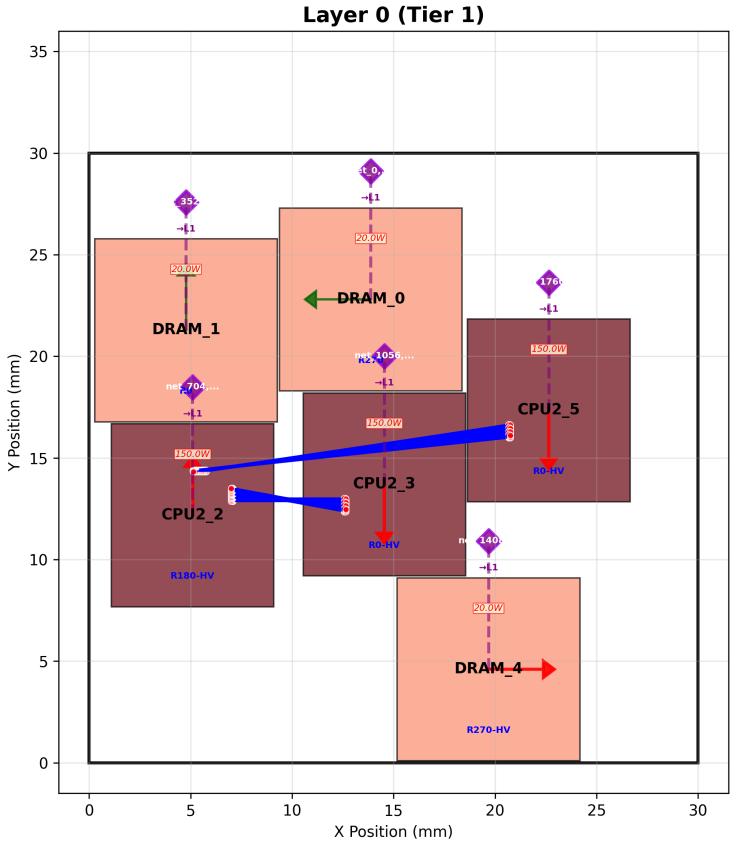


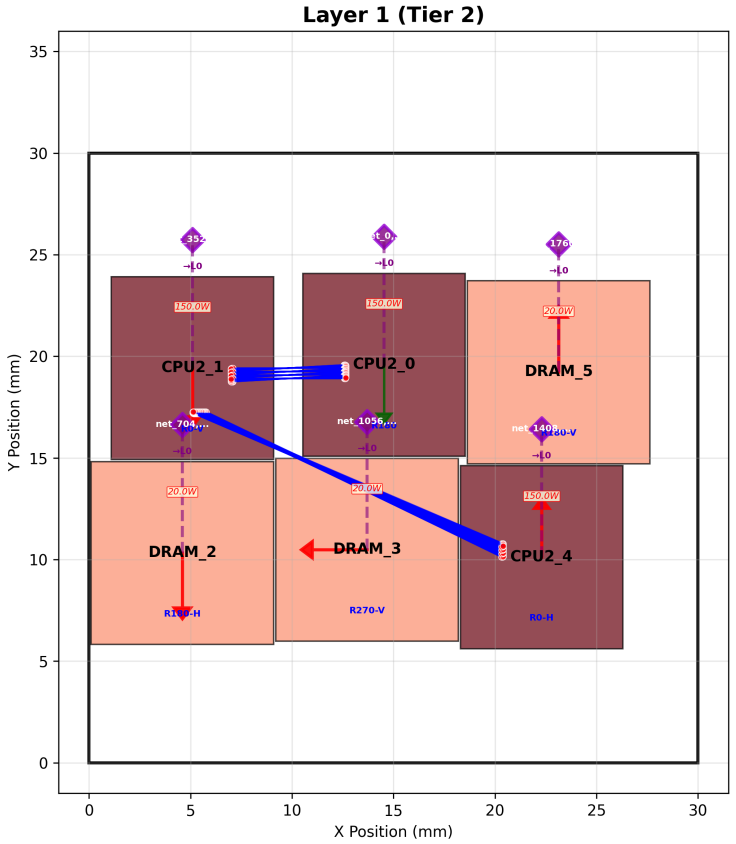


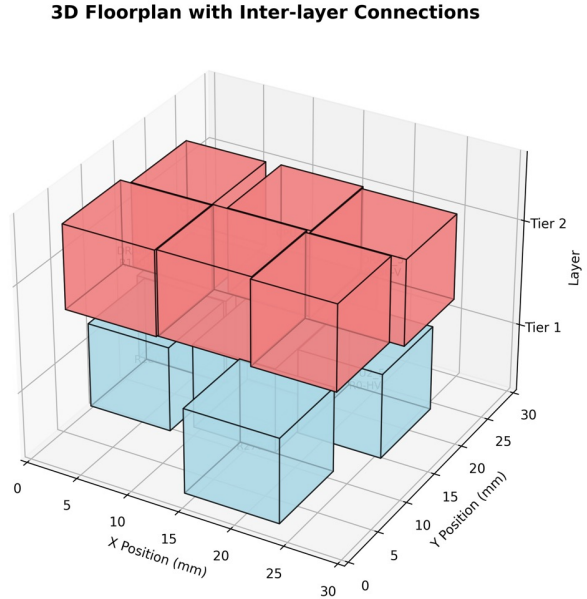



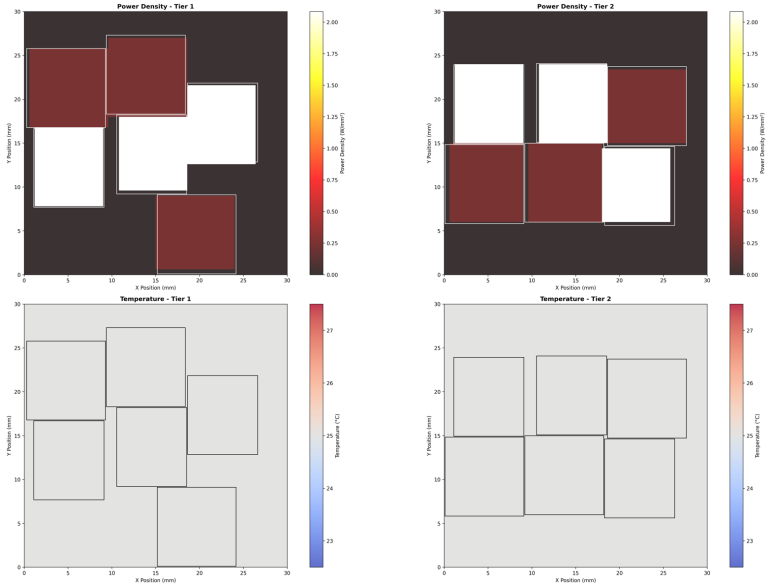

Time: 62.58
Peak Temperature: 317.46
HPWL: 10147.99 mm

# Placer Experiments – Case 4



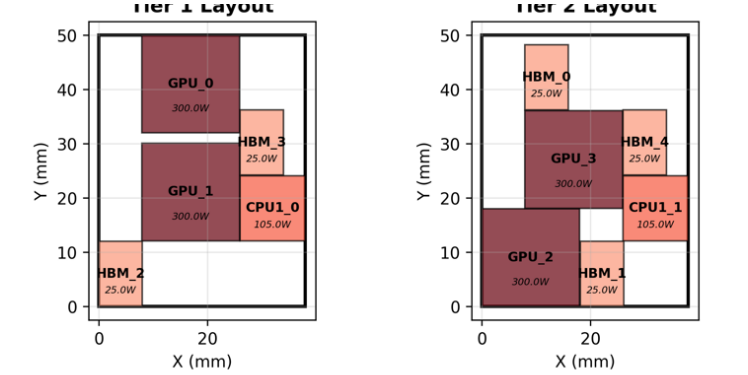



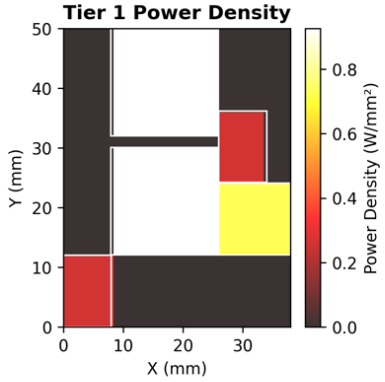

Time: 1202.22
Peak Temperature: 311.83
HPWL: 16995.43 mm

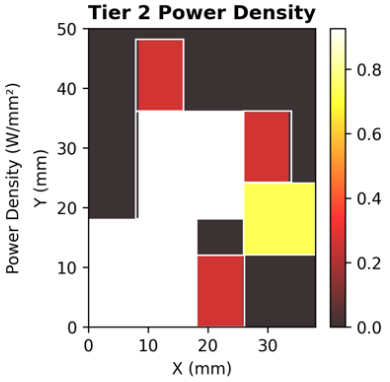
# Placer Experiments – Case 5

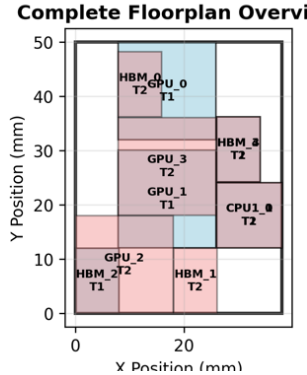








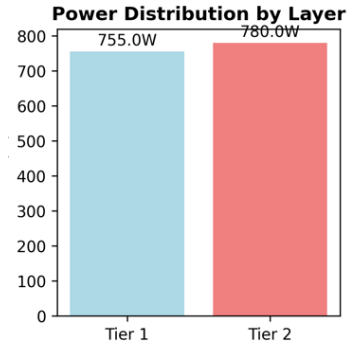



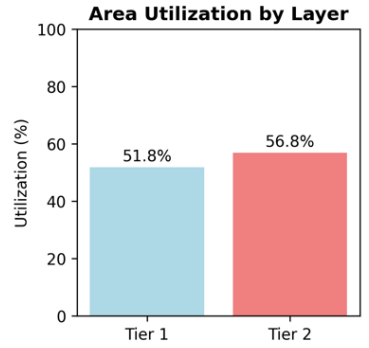



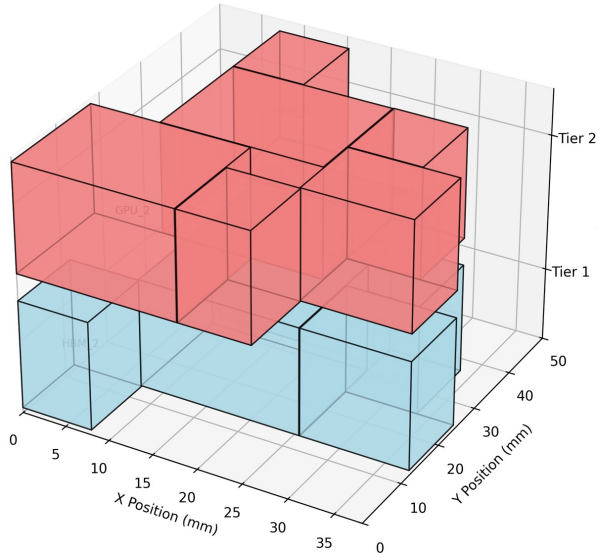

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 11
Total Nets: 3520
Layers: 2

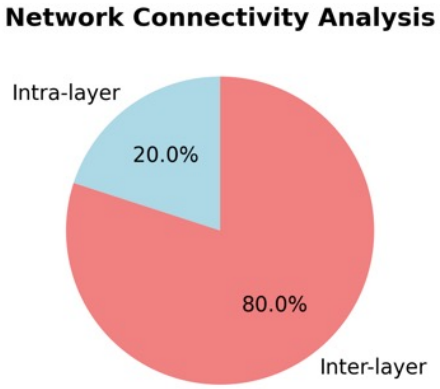
COSTS:
Total: 31221.92
Wirelength: 31171.92
Thermal: 25.00

POWER:
Total: 1535.0 W
Tier 1: 755.0 W
Tier 2: 780.0 W
```



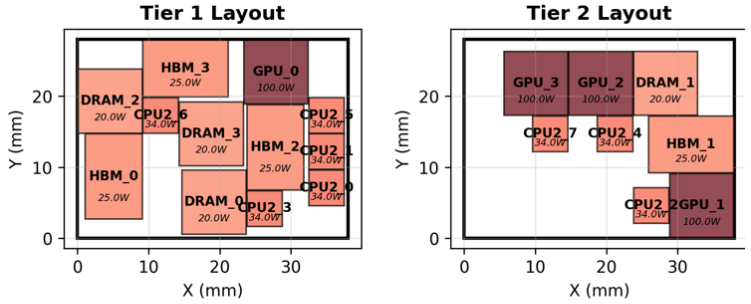



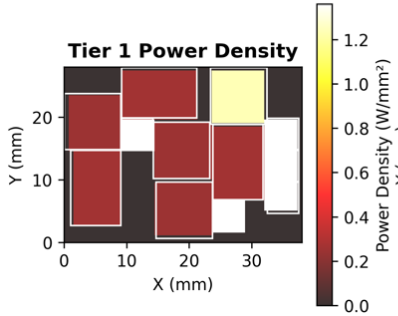

Peak Temp.315.2

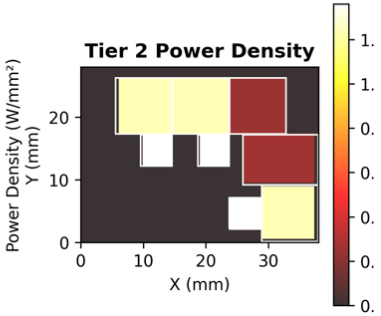
# Placer Experiments – Case 6

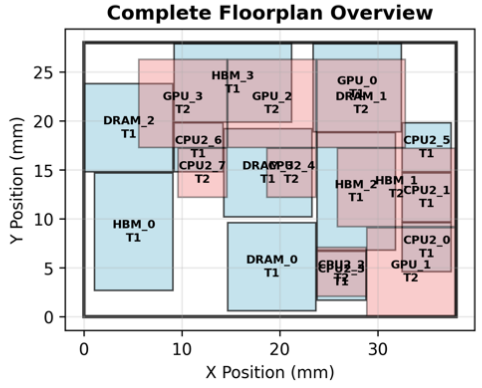








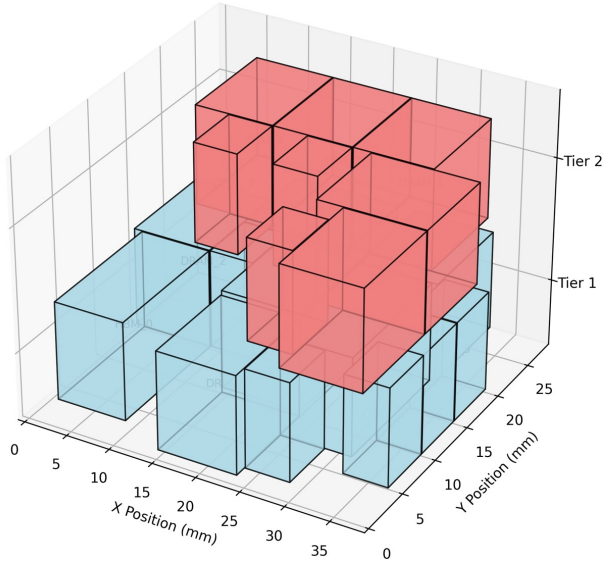



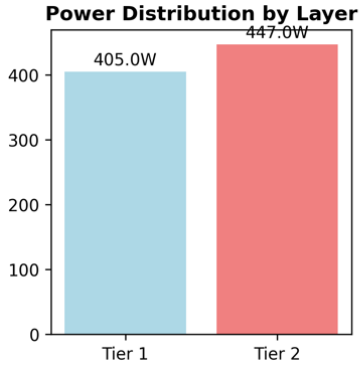



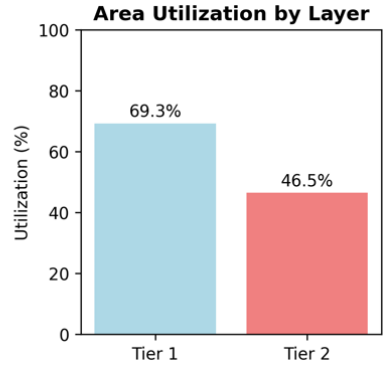

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 20
Total Nets: 2816
Layers: 2

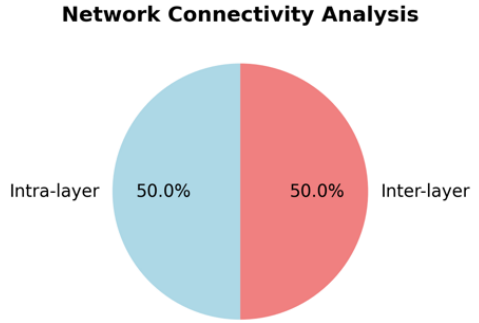
COSTS:
Total: 22488.02
Wirelength: 22438.01
Thermal: 25.00

POWER:
Total: 852.0 W
Tier 1: 405.0 W
Tier 2: 447.0 W
```



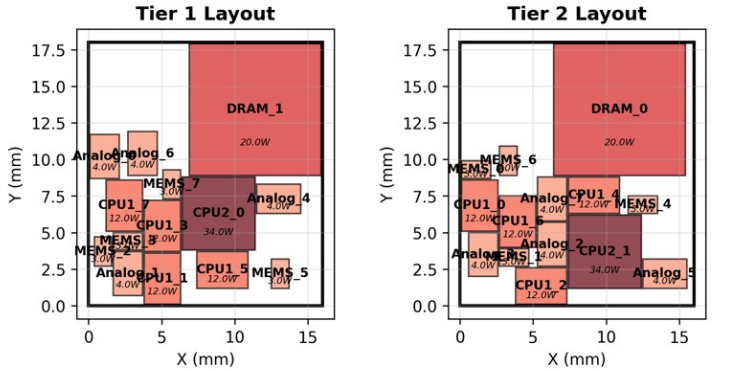



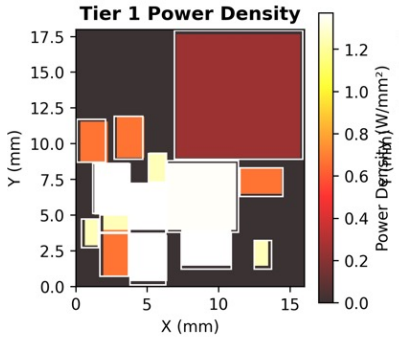

Time: 69.69
Peak Temp.: 283.39
HPWL: 22438.01 mm

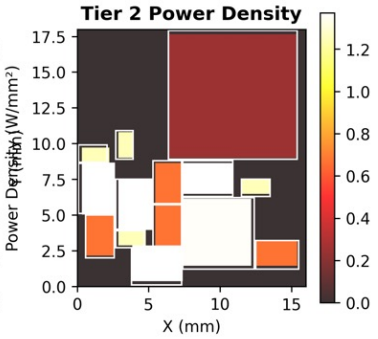
# Placer Experiments – Case 7

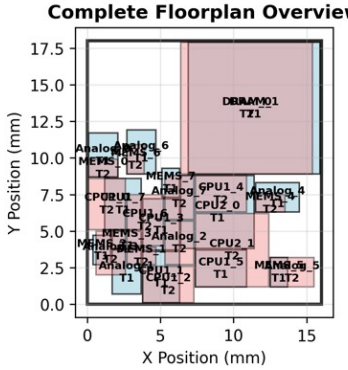








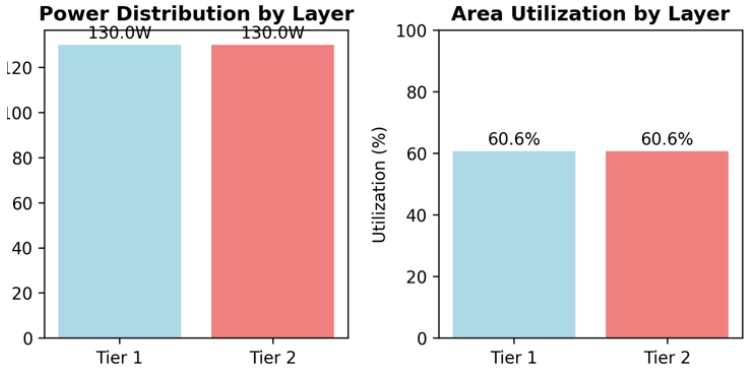



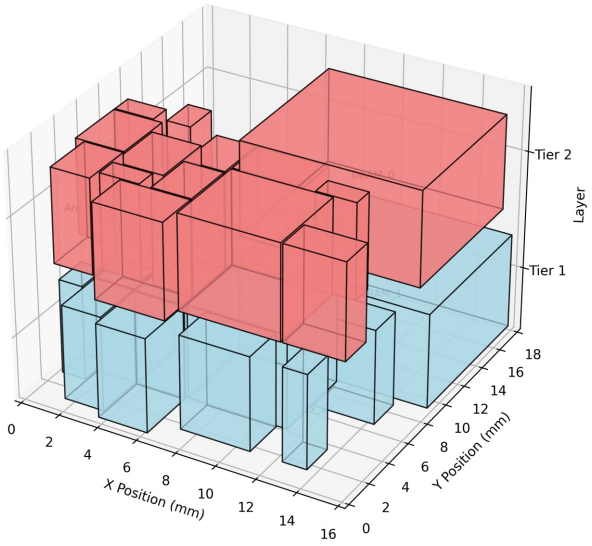



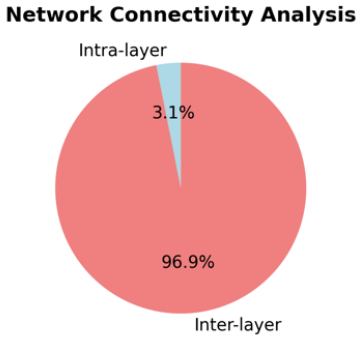

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 28
Total Nets: 1408
Layers: 2

COSTS:
Total: 4289.54
Wirelength: 4239.53
Thermal: 25.01

POWER:
Total: 260.0 W
Tier 1: 130.0 W
Tier 2: 130.0 W
```

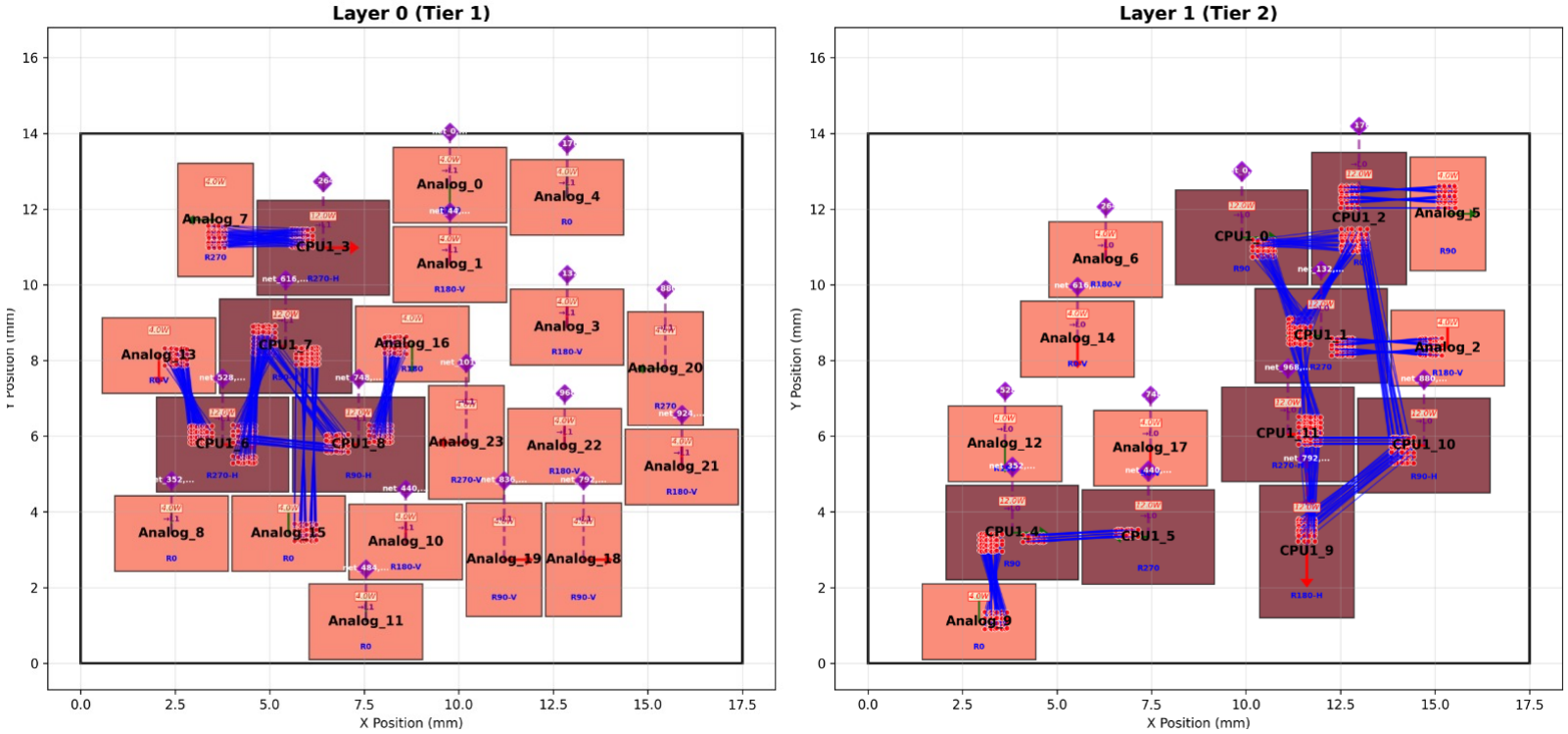


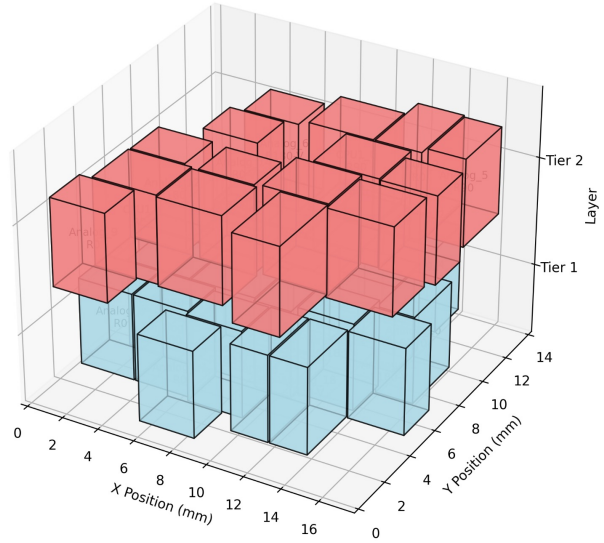




Time: 71.87
Peak Temp.: 303.32
HPWL: 4289.54 mm

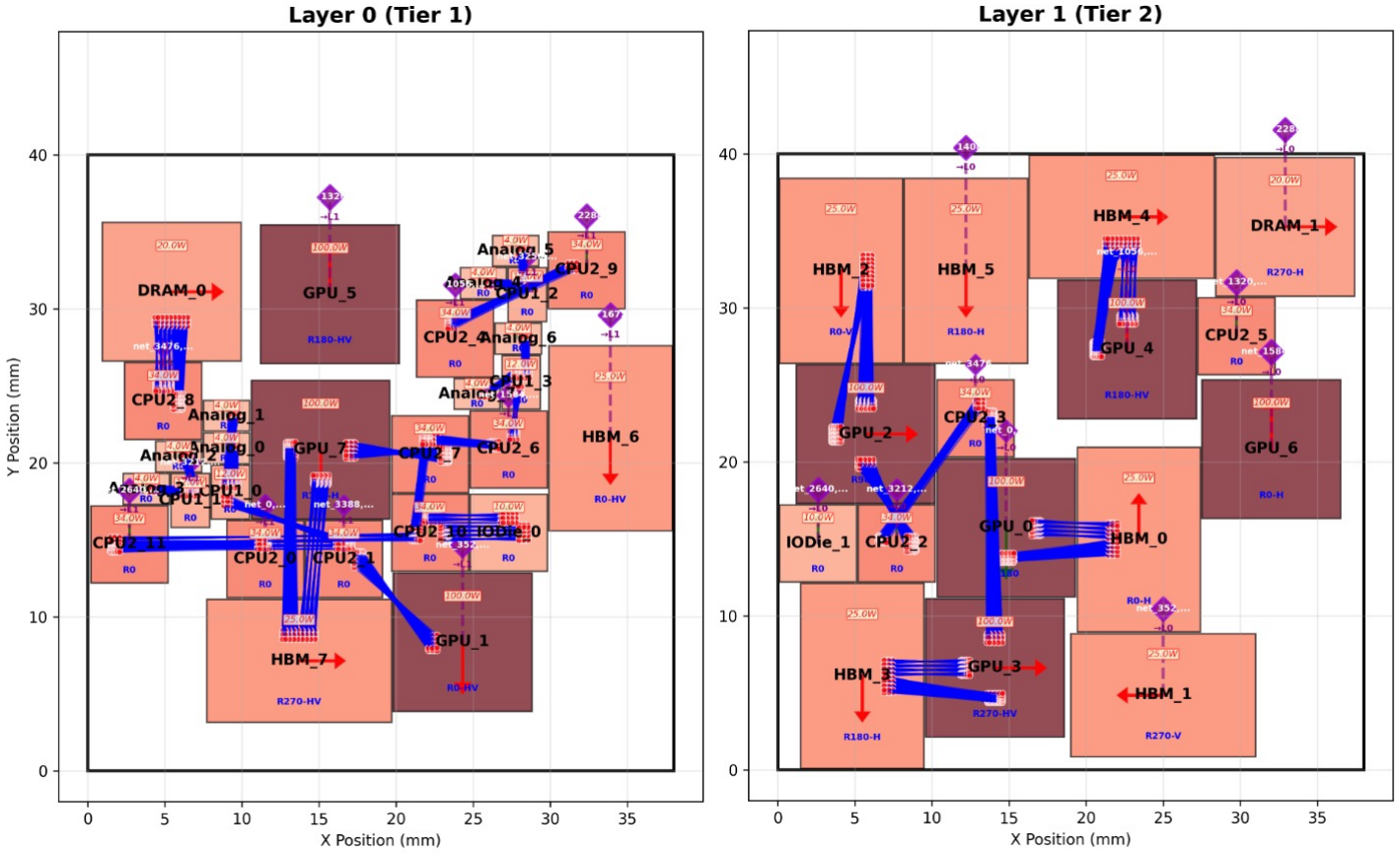
# Placer Experiments – Case 8

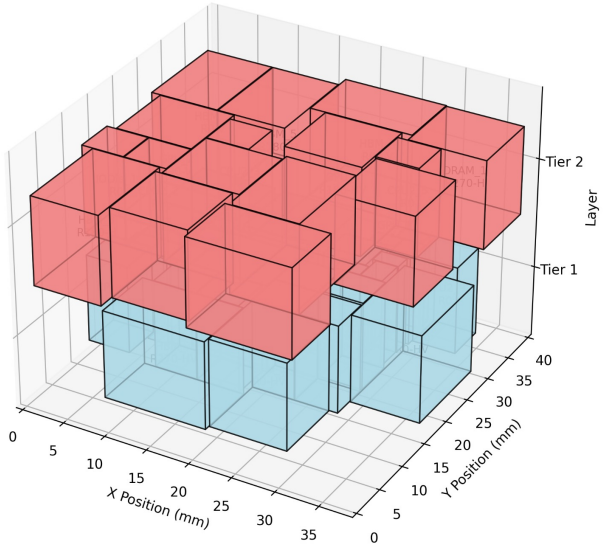



Time 3504.624584913254
Peak temp. 282.85
HPWL: 3361.93 mm

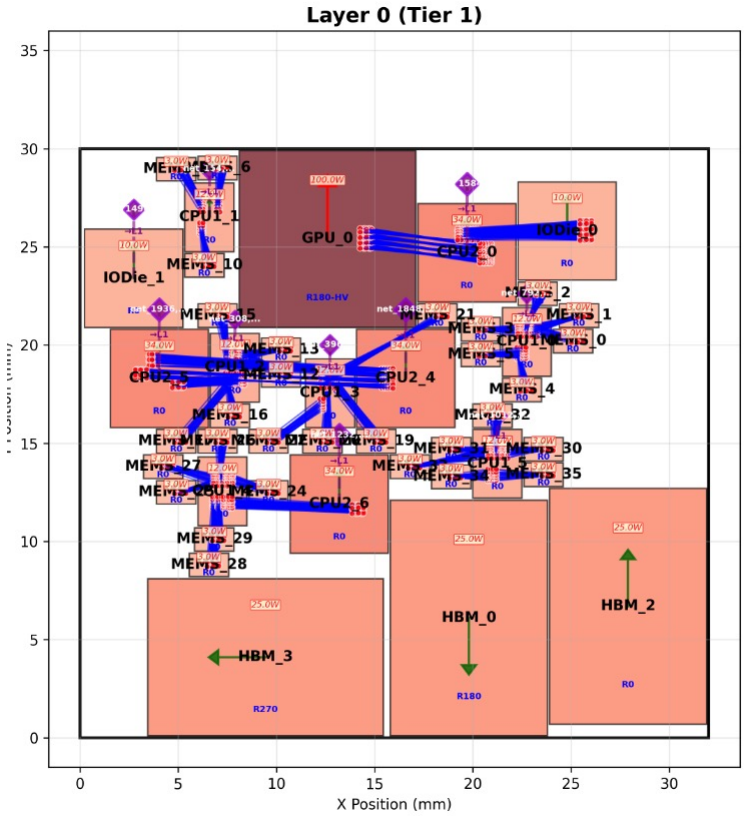
# Placer Experiments – Case 9

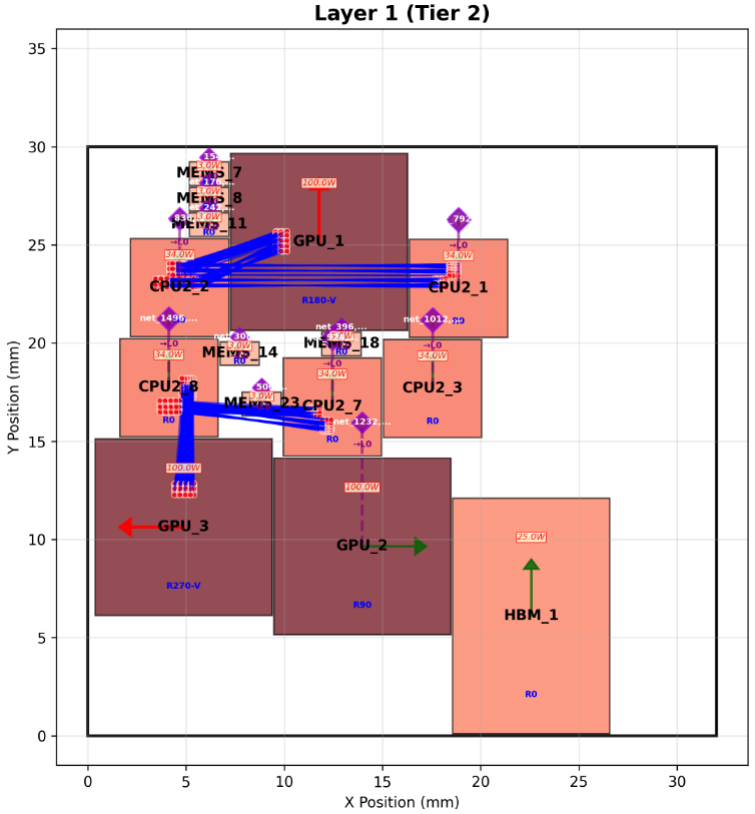


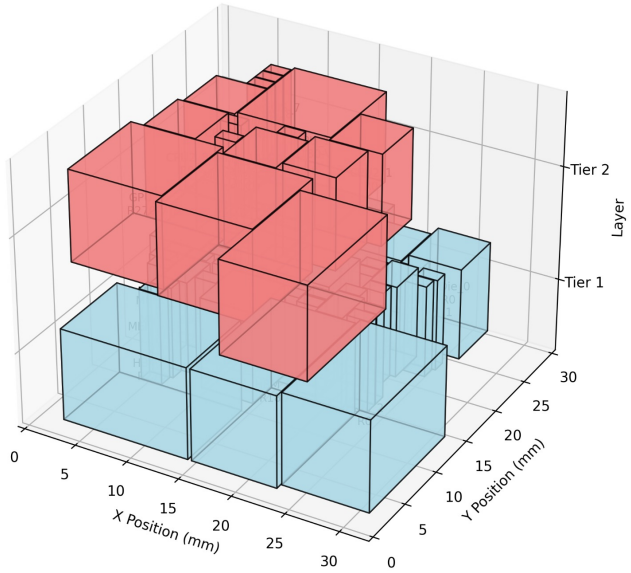



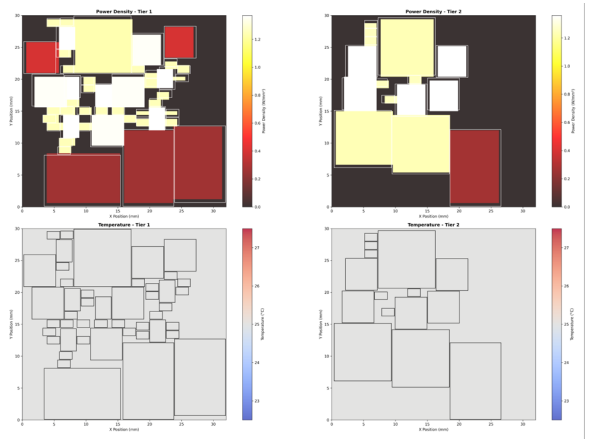

Time 3507.0527629852295
Peak temp. 334.82
HPWL: 48358.59 mm

# Placer Experiments – Case 10



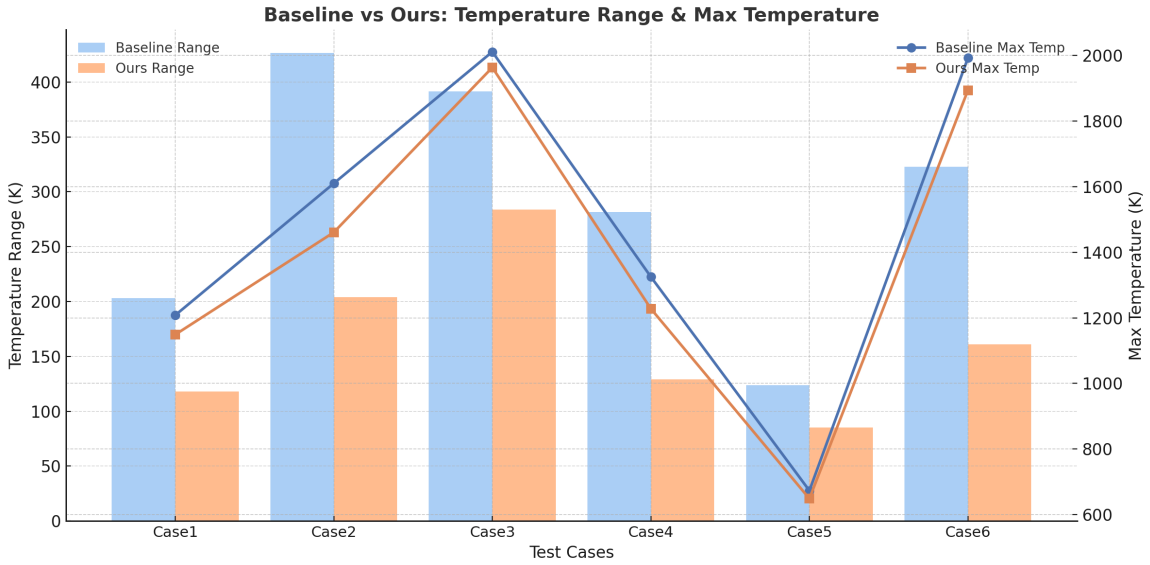



Time 3507.3127374649
Peak temp. 300.15
HPWL: 28138.81 mm

# Comparison Experimental Results – Temp. Peak

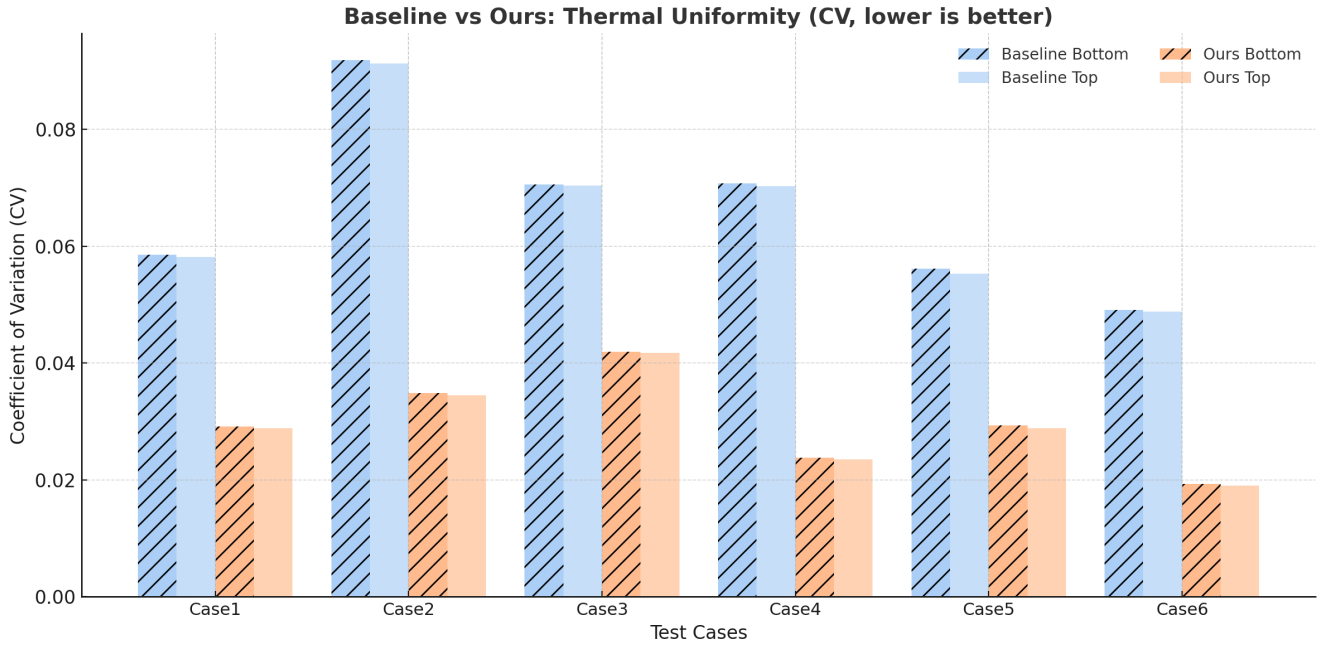



For each test case, our method get “better” results:
- Lower Max. Temperature;
- Smaller Temperature Range.

On average, peak temperature decreases by ~17K (≈1.2%), and in Case 6 our method achieves a substantial reduction of nearly 100K compared with the baseline.

| Ours | Bottom Die (K) | | | Top Die (K) | | | Baseline | Bottom Die (K) | | | Top Die (K) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Min. | Max. | Range | Min. | Max. | Range | | Min. | Max. | Range | Min. | Max. | Range |
| 1 | 1030.89 | 1148.89 | 118 | 1030.86 | 1147.73 | 116.87 | 1 | 1030.89 | 1148.89 | 118 | 1030.86 | 1147.73 | 202.06 |
| 2 | 1256.43 | 1460.58 | 204.15 | 1256.36 | 1457.89 | 201.53 | 2 | 1256.43 | 1460.58 | 204.15 | 1256.36 | 1457.89 | 423.94 |
| 3 | 1679.34 | 1963.14 | 283.8 | 1679.31 | 1961.91 | 282.6 | 3 | 1679.34 | 1963.14 | 283.8 | 1679.31 | 1961.91 | 390.13 |
| 4 | 1098.67 | 1227.89 | 129.22 | 1098.65 | 1226.17 | 127.52 | 4 | 1098.67 | 1227.89 | 129.22 | 1098.65 | 1226.17 | 280.05 |
| 5 | 563.78 | 648.94 | 85.16 | 563.76 | 647.44 | 83.68 | 5 | 563.78 | 648.94 | 85.16 | 563.76 | 647.44 | 122.23 |
| 6 | 1732.74 | 1893.58 | 160.84 | 1732.71 | 1890.03 | 157.32 | 6 | 1669.33 | 1991.97 | 322.64 | 1669.31 | 1990.27 | 320.96 |

# Experimental Results – Temp. Balanced Distribution

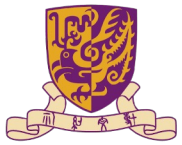



We use Coefficient of Variation (CV) to evaluate the balanced or not of the die temperature distribution.

$$CV = \frac{\sigma}{\mu}$$

Compared with the baseline, our method improves thermal uniformity by reducing the coefficient of variation (CV) by approximately 55.1%.

| | Ours | | | | Baseline | | | |
|---|---|---|---|---|---|---|---|---|
| | Bottom Die | | Top Die | | Bottom Die | | Top Die | |
| Test case | Mean. | Std. | Mean. | Std. | Mean. | Std. | Mean. | Std. |
| 1 | 1079.874 | 31.49698 | 1079.466 | 31.16869 | 1080.114 | 63.22427 | 1079.611 | 62.75536 |
| 2 | 1310.281 | 45.66117 | 1309.928 | 45.18138 | 1310.527 | 120.3783 | 1309.996 | 119.519 |
| 3 | 1788.517 | 75.05186 | 1788.027 | 74.68629 | 1788.06 | 126.1314 | 1787.677 | 125.717 |
| 4 | 1143.345 | 27.165 | 1142.97 | 26.86789 | 1143.296 | 80.84383 | 1142.92 | 80.35285 |
| 5 | 587.1468 | 17.23045 | 586.723 | 16.92625 | 587.2032 | 32.95235 | 586.7746 | 32.44569 |
| 6 | 1806.552 | 34.86248 | 1805.685 | 34.31764 | 1806.163 | 88.57475 | 1805.684 | 88.1785 |



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# Thanks for Your Attention!